

TEACHME-LAKEHOUSE · TRAINING PROGRAMME

01

Redshift Hands-On Labs

Four days of AWS Redshift labs, mapped lesson by lesson

COMPANION TO MODULE 01

REDSHIFT DEEP DIVE

4 DAYS

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ISSUED

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How to use this. Module 01 is 45 lessons of *why*. This is the *doing*. Each day pairs a block of lessons with a real AWS lab that provisions real infrastructure.

Read the lessons first, then do the lab. The lab makes it stick; the lesson makes the lab make sense. Doing the labs cold is how people finish a workshop having clicked forty buttons and learned nothing.

Link verification

Every URL below was fetched and confirmed live on **12 August 2026**. Two corrections to note before you send anyone a link:

RESOURCE	STATUS	WHAT I CONFIRMED
Redshift Immersion Day	⚠ use the canonical URL	The old redshift-immersion.workshop.aws returns 301 Moved Permanently to a UUID URL. Both work today, but cite _catalog.workshops.aws/redshift-immersion/en-US — the UUID form is a redirect <i>target</i> and can change
Data Engineering Immersion Day	✅ active · 198★ · 119 commits	Lab titles confirmed verbatim: Lab 1 "Hydrating the data lake via DMS", Lab 6 "Modernize Data Warehouse with Amazon Redshift Spectrum". Has an AutoComplete DMS shortcut if Lab 1 setup runs long
dbt + Redshift pipelines	✅ live	Full title: "Build and Deploy Data Pipelines with Amazon Redshift, dbt...". Uses dbt and MWAA
Redshift Streaming Workshop	✅ active, but 13★ · 33 commits	Smallest and least-trafficked of the four. CDK (Python) provisions Kinesis, Redshift, Lambda, Step Functions, SNS and Managed Grafana . Also includes a Redshift ML step
MSK streaming best practices	✅ live · 11 Mar 2024	Authors Poulomi Dasgupta, Adekunle Adedotun. Contains the IAM policies (incl. cross-account) and the <code>CREATE MATERIALIZED VIEW ... WHERE CAN_JSON_PARSE(kafka_value)</code> pattern, plus an incremental-load stored procedure using Kafka partition/offset as the CDC marker

Redshift Immersion Day contains **more than 23 labs**. It cannot be done in a day, and attempting Levels 100–300 in two days means skimming. **Select deliberately** — the day plans below name the topics to pick, not "do the workshop".

The four-day plan

I have restructured this from the three days originally proposed, for two reasons: the lab count above, and because **streaming ingestion is not Module 01 material** — it is taught in Module 00. Day 4 is therefore an extension day, not a core day.

Pick the labs covering: cluster/workgroup creation · loading with `COPY` · **table design and distribution/sort keys** · compression.

The table-design lab is the single most valuable hour in the whole track. It is L14–L20 made physical: you build the same table two ways and watch the query time change.

Do this during the lab: after each design change, run the `svv_table_info` query from L43 and record `size` , `skew_rows` , `unsorted` and `stats_off` . By the end of the day the team should read that view without prompting.

TPC benchmark load, then [Data Engineering Immersion Day](#)

The TPC load gives them `COPY` at a scale where design mistakes actually hurt. Lab 1 hydrates a data lake via **DMS** — which is exactly the Tamimi and Apparel Group source pattern. Lab 6 is Spectrum, i.e. L12 made real.

Use the AutoComplete DMS shortcut if Lab 1 setup overruns. Getting to Lab 6 matters more than watching DMS provision.

[Immersion Day](#) WLM / performance labs, **plus the two custom labs below**

This is the day with the biggest gap in AWS's material, and the day that most changes how the team works. See **the gap** section next.

[dbt + Redshift](#), then [Streaming](#)

Do **dbt** in the morning: it is SQL transformation with tests and lineage, and it maps onto L31–L35. If MWAA cost or setup time is a concern, use [dbt CLI and Amazon Redshift](#) instead — same dbt, no orchestration layer.

Do **streaming** in the afternoon only if the team is ahead — it is Module 00 material and provisions the most expensive stack in this document. Pair it with the [MSK best-practices blog](#) for the IAM policies and the materialized-view SQL.

The gap: two lessons no AWS lab covers

I checked all four workshops against the 45 lessons. **The two lessons written specifically for this team have no lab anywhere in AWS's catalogue:**

- **L39 · Calling Redshift From Node.js** — every workshop uses Query Editor v2. None of them writes application code against the Data API.
- **L44 · Diagnosing A Slow Query** — the workshops show you a fast query. None hands you a slow one and asks why.

These are the two most important lessons for three Node.js developers becoming warehouse engineers, so they need purpose-built exercises. Both are already specified in the take-homes; run them as timed labs on **Day 3**.

From L39 "Try it" · run against the Day 1 cluster

1. Write the Data API client: `execute` with exponential backoff, `query` with `NextToken` pagination and typed-union unwrapping.
2. Return 5,000 rows and confirm you get all of them. **Then delete the `NextToken` loop and count what you get.** That silent truncation is the exercise.
3. Load 10,000 rows twice — a loop of `INSERT` s, and S3 + `COPY` . Time both, then compare `size` and `unsorted` in `svv_table_info` . **Put both block counts on the board.**
4. Move a three-statement job into one `CALL` and prove the failure rolls back cleanly.
5. Switch from a stored password to `GetCredentials` temporary credentials.

Step 3 is the one that permanently changes how they write load code.

From L44 · use the TPC tables from Day 2

1. Deliberately break a good query four ways — skip the `ANALYZE` , set the small dimension to `DISTSTYLE EVEN` , widen a `VARCHAR` to 65535, wrap the sort column in `DATE_TRUNC` .
2. One person drives; everyone else works the six questions **in order** and writes down each answer before moving on.
3. **Whoever suggests rewriting the SQL before question 6 buys the coffee.**
4. Fix, re-measure with the result cache off, confirm with `EXPLAIN` .
5. Write the diagnosis in a one-paragraph note in the repo.

Four induced faults, six questions, four correct diagnoses. If the team can do this, they can operate the warehouse.

Lesson → lab coverage

Where each part of Module 01 gets hands-on reinforcement:

PART	LESSONS	LAB COVERAGE
A · What Redshift is	L01–L07	✅ Immersion Day Level 100
B · Objects and types	L08–L13	● Partial — Spectrum well covered (Lab 6), <code>SUPER</code> only in the streaming lab
C · Physical design	L14–L20	✅ Strong — the table-design lab is the best in the track
D · Getting data in and out	L21–L27	✅ Strong — TPC load + DMS Lab 1
E · SQL that performs	L28–L34	● Partial — Custom Lab B fills the diagnosis gap
F · Programmability	L35–L40	● dbt covers transformation · Custom Lab A fills the Node.js gap
G · Operating it	L41–L45	● WLM covered · no cost or maintenance lab — use the L42/L45 queries on the live cluster

testing and lineage model, Airflow orchestration, Kinesis streaming ingestion, Redshift ML, and Managed Grafana dashboards. Worth knowing they are there.

⚠ Cost and teardown

These labs provision **real, billable infrastructure**. Two of them are expensive and easy to leave running:

LAB	WATCH OUT FOR
Streaming workshop	Managed Grafana and the Kinesis stream bill continuously. Run <code>cdk destroy</code> the same day — it is in the lab, do not skip it
dbt + MWAA	MWAA is the most expensive component in this document and does not auto-pause
Data Engineering Day	DMS replication instances keep running after the lab ends
Immersion Day	Use a Serverless workgroup where the lab allows it — it auto-pauses overnight; a provisioned cluster does not

one person is named teardown owner and confirms in writing. A forgotten MWAA environment over a weekend costs more than the whole training.

Prerequisites

- A **dedicated training AWS account** — never the Tamimi Dev account. These labs create and delete infrastructure freely.
- Elevated IAM privileges (the CDK and CloudFormation stacks need them).
- Locally: **CDK**, **Docker CLI**, **Python 3**, **Node 18+**, `git`, `psql` and the AWS CLI v2 with SSO configured.
- Access to Query Editor v2 in the console.
- Module O1 take-homes open in a second window. The SQL in them is the lab notebook.

Every link, in one place

#	RESOURCE	URL
1	Redshift Immersion Day (canonical)	https://catalog.workshops.aws/redshift-immersion/en-US
2	Data Engineering Immersion Day (GitHub)	https://github.com/aws-samples/data-engineering-for-aws-immersion-day
3	Build and Deploy Data Pipelines with Redshift + dbt	https://catalog.workshops.aws/opensource-with-redshift/en-US
4	dbt CLI and Amazon Redshift (gentler starting point)	https://catalog.workshops.aws/dbt-cli-and-amazon-redshift/en-US/introduction
5	Amazon Redshift Streaming Workshop (GitHub)	https://github.com/aws-samples/amazon-redshift-streaming-workshop
6	Near-real-time analytics with MSK (blog)	https://aws.amazon.com/blogs/big-data/best-practices-to-implement-near-real-time-analytics-using-amazon-redshift-streaming-ingestion-with-amazon-msk/
7	Awesome AWS Workshops (index)	https://awesome-aws-workshops.com/

runs short or if MWAA cost is a concern; it teaches dbt itself without the orchestration infrastructure.

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